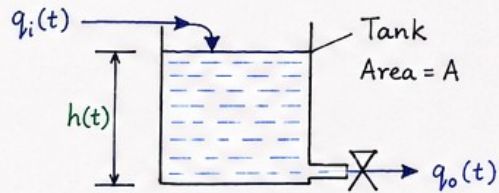


ASSIGNMENT - INDUSTRIAL CONTROL (PRACTICAL QUESTIONS)

Solve any FOUR questions. Show all steps clearly.

Q1. INDUSTRIAL CONTROL EXAMPLE & MATHEMATICAL MODEL

Consider a level control system of a single tank as shown in the figure.



$q_i(t)$: Inlet flow rate

$q_o(t)$: Outlet flow rate

$h(t)$: Liquid level

Assume linear relation $q_o(t) = (1/R) h(t)$, R - flow resistance.

Derive the mathematical model relating $H(s)$ and $Q_i(s)$.

Hence, obtain the transfer function

$$G_p(s) = \frac{H(s)}{Q_i(s)} = \frac{R}{ARs + 1}$$

Q2. CONTROL HARDWARE & THEIR MODELS

(a) Draw the symbol and give the mathematical model of the following control hardware.

i) DC Servo Motor ii) Tachogenerator iii) Potentiometer

(b) A DC servo motor has the following specifications:

$J = 0.01 \text{ kg} \cdot \text{m}^2$, $B = 0.1 \text{ N} \cdot \text{m} \cdot \text{s}/\text{rad}$, $K_t = 0.01 \text{ N} \cdot \text{m}/\text{A}$,

$R_a = 1 \Omega$, $L_a = 0.5 \text{ H}$, $K_b = 0.01 \text{ V} \cdot \text{s}/\text{rad}$.

Derive the transfer function $\frac{\Omega(s)}{V_a(s)}$ for the motor.

(c) What is the function of each hardware in a control system?

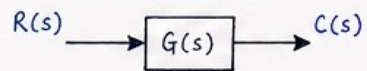
Write any two practical applications of each.

Note: Assume zero initial conditions.

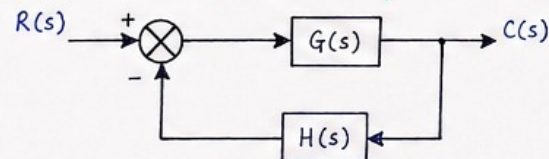
Q3. FEEDBACK CONTROL: OPEN-LOOP & CLOSED-LOOP SYSTEMS

The position control system of a motor in open-loop and closed-loop forms are shown below.

Open-Loop System



Closed-Loop System (Negative Feedback)



(a) Derive the closed-loop transfer function $\frac{C(s)}{R(s)}$.

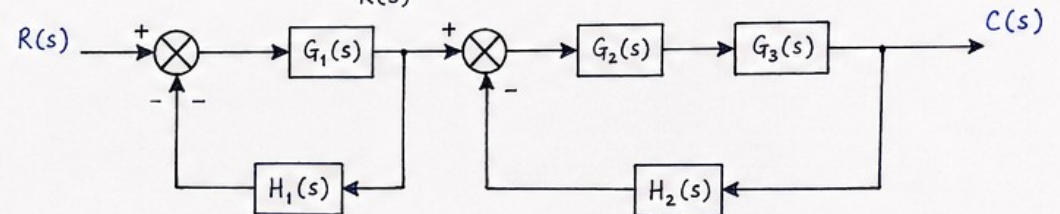
(b) List any FOUR benefits of feedback control over open-loop control.

(c) For unity feedback ($H(s) = 1$) and $G(s) = \frac{10}{s(s+2)}$, determine the steady-state error for a unit step input.

Note: Assume stable system.

Q4. BLOCK DIAGRAM ALGEBRA

Reduce the following block diagram to a single equivalent transfer function $\frac{C(s)}{R(s)}$.



Given:

$$\begin{aligned} G_1(s) &= \frac{2}{s+1}, & G_2(s) &= \frac{5}{s}, & G_3(s) &= \frac{1}{s+2} \\ H_1(s) &= 0.5, & H_2(s) &= \frac{1}{s+3} \end{aligned}$$

Find $\frac{C(s)}{R(s)}$ and hence comment on the type of system.